

Lab 20 - Troubleshoot a Wireless Connection

Objectives and Background

Identify and correct any misconfiguration of a wireless device.

A small business owner learns that a wireless user is unable to access the network. All the PCs are configured with static IP addressing. Identify and resolve the issue.

Instructions

Part 1: Verify connectivity.

Access the Desktop > Web Browser of each wireless PC and type `www.cisco.pka` into the URL. Identify any PCs that are not connecting to the web server.

Note: All the devices require time to complete the boot process. Please allow up to one minute before receiving a web response.

Question:

Which wireless PCs are unable to connect to the web server?

Part 2: Examine the IP configuration of the PCs.

- a. On the PC that is unable to connect, access the Command Prompt from the Desktop tab.
- b. Type the `ipconfig /all` command.

Question:

What IP addressing information is available? `Apiipa, failed DHCP 169.254.122.96, Self assigned`

Part 3: Examine the Wireless Settings on the Wireless Client.

- a. On the Desktop tab of any PC that is unable to connect, click PC Wireless to access the wireless configurations.

Question:

b. Click Connect tab and record the associated SSID. Click Refresh as needed to display the list of SSID.

Part 4: Examine the Wireless Settings on the Wireless Router.

a. Access the Wireless Router from the web browser of a wired PC. Use the username admin and password admin to access the wireless router.

Question:

What IP address did you use? (Hint: default gateway) DG: 192.168.1.1

c. On the Basic Setup page, examine the DHCP Server Setting configuration.

Question:

Is DHCP enabled? Yes

d. Click the Wireless tab.

e. Examine the setup information under the Wireless tab.

Question:

What is the SSID? Does it match the SSID configured on the client? Academy, Yes

f. Click the Wireless Security submenu.

g. Examine the security settings.

Question:

What is the wireless security mode? What is the passphrase? WPA2-Personal, Cisco123

Part 5: Make any necessary configuration changes on the Wireless Clients

a. On the Desktop tab of any PC that is unable to connect. Click PC Wireless to correct the wireless configurations.

b. Click Connect tab. Select the Academy wireless network and click Connect.

c. Enter the passphrase (Pre-shared Key) recorded from the wireless router. Click Connect.

d. Using the Web Browser within the Desktop tab connect to www.cisco.pka to verify that the configuration changes resolved the problem.

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